Assignment No : 07

1. Title

Case Study on Microsoft Azure: Deployment of a Static Web Application

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2. Problem Statement

Microsoft Azure is a cloud computing platform and infrastructure created by Microsoft for building, deploying, and managing applications and services through a global network of Microsoft-managed data centers.

The objective of this case study is to understand Azure services and deploy a static website using Azure cloud infrastructure.

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3. Objectives

- To understand the concept of **cloud computing**
- To study the features of Microsoft Azure
- To learn how to deploy a **static web application**
- To gain practical experience with Azure portal
- To understand cloud-based hosting and scalability

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4. Introduction to Microsoft Azure

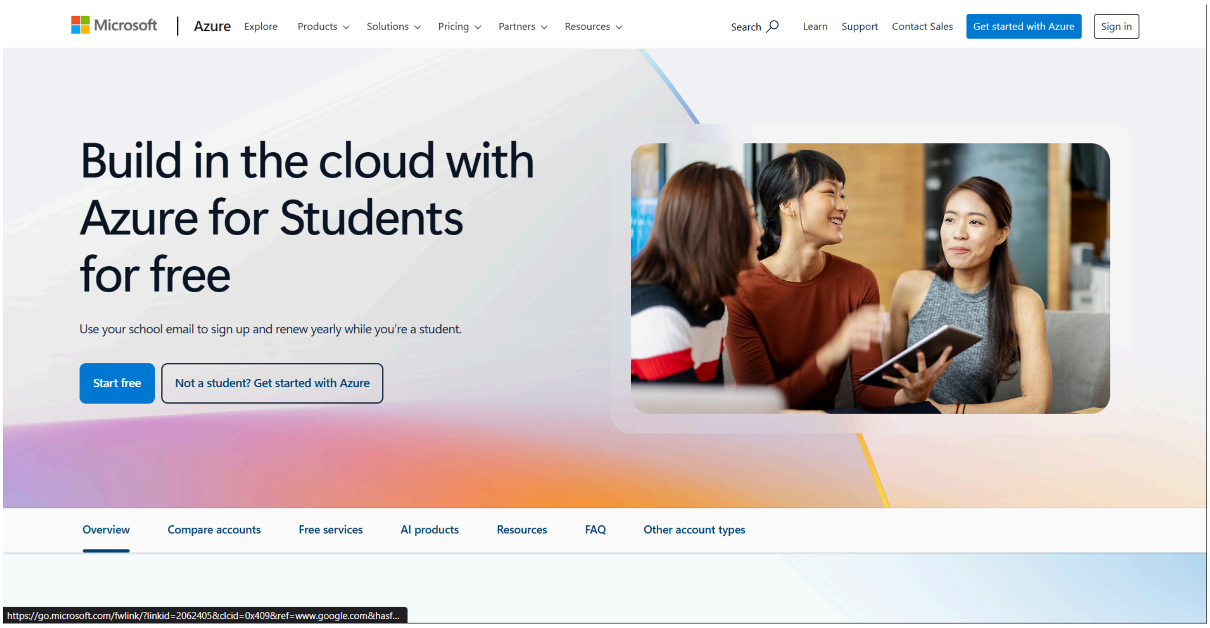
Microsoft Azure is a cloud computing platform that provides services such as:

- Virtual Machines
- Storage
- Networking
- Databases
- AI & Analytics
- Web Hosting

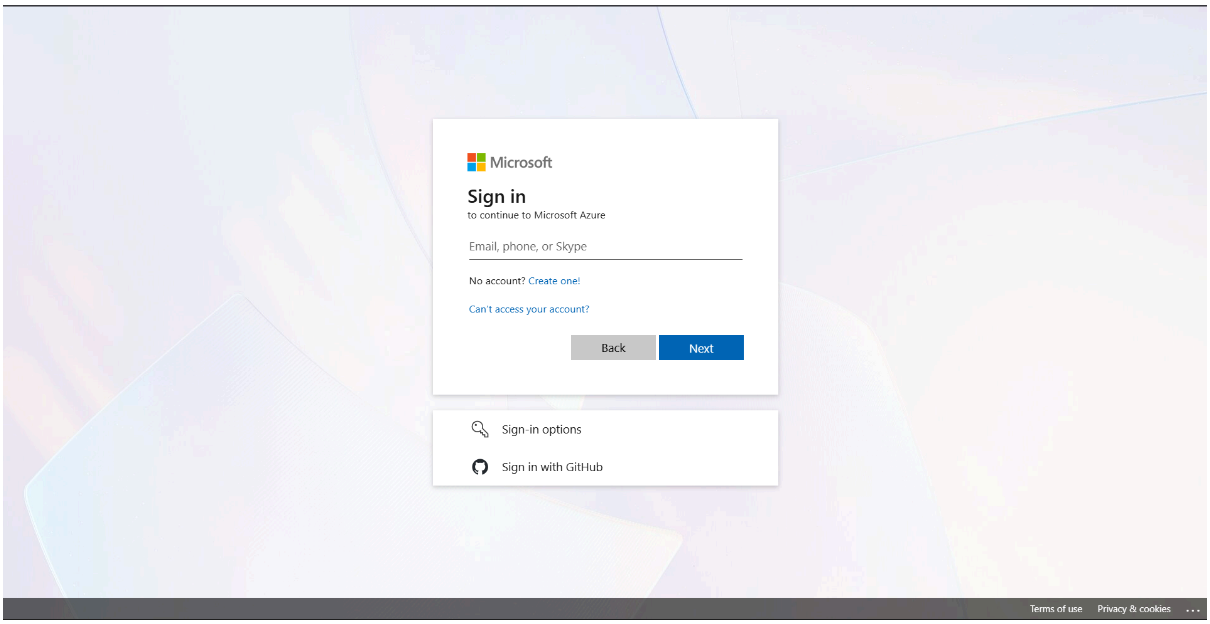
Steps to Perform: Hosting Static Website on Azure

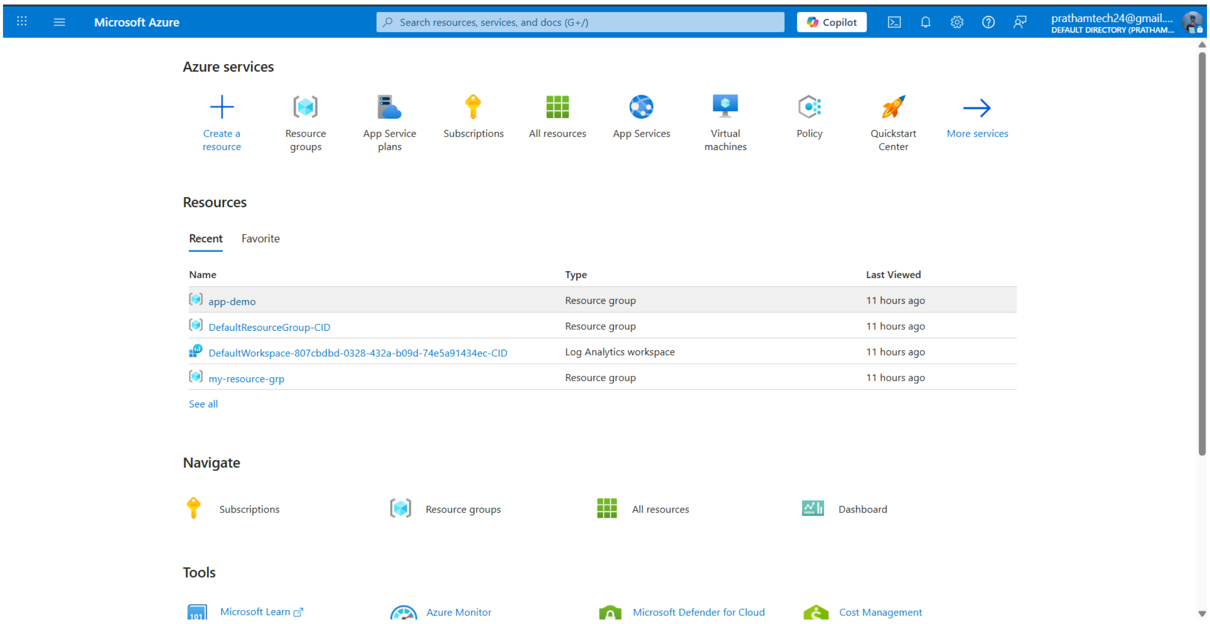
Deployment Method: Azure Static Web Apps (GitHub Integration)

1. Account Setup and Authentication

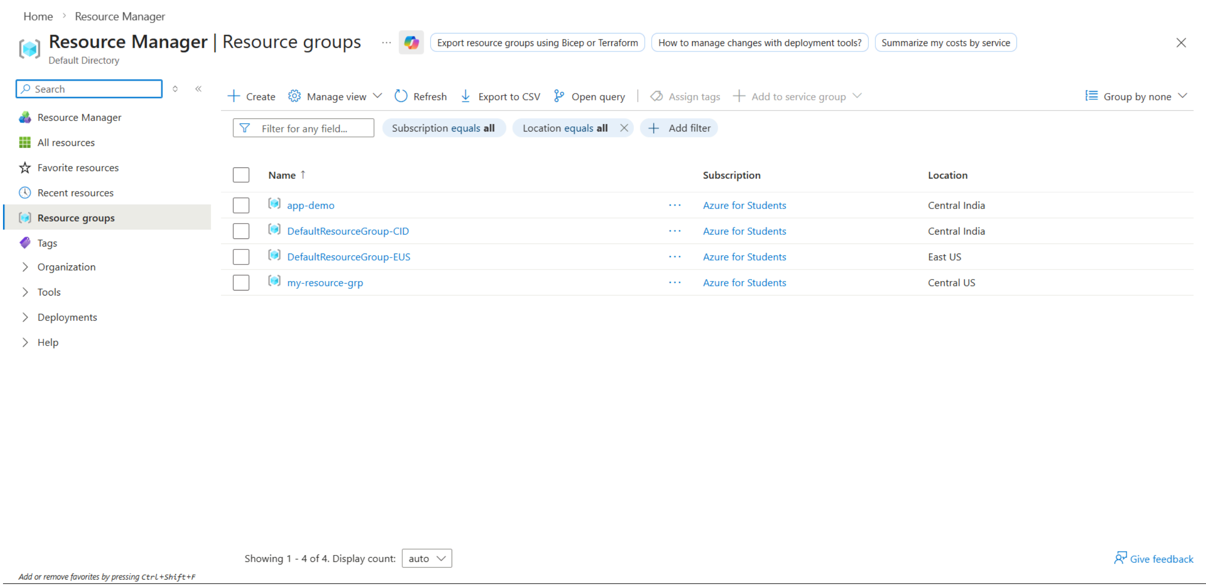


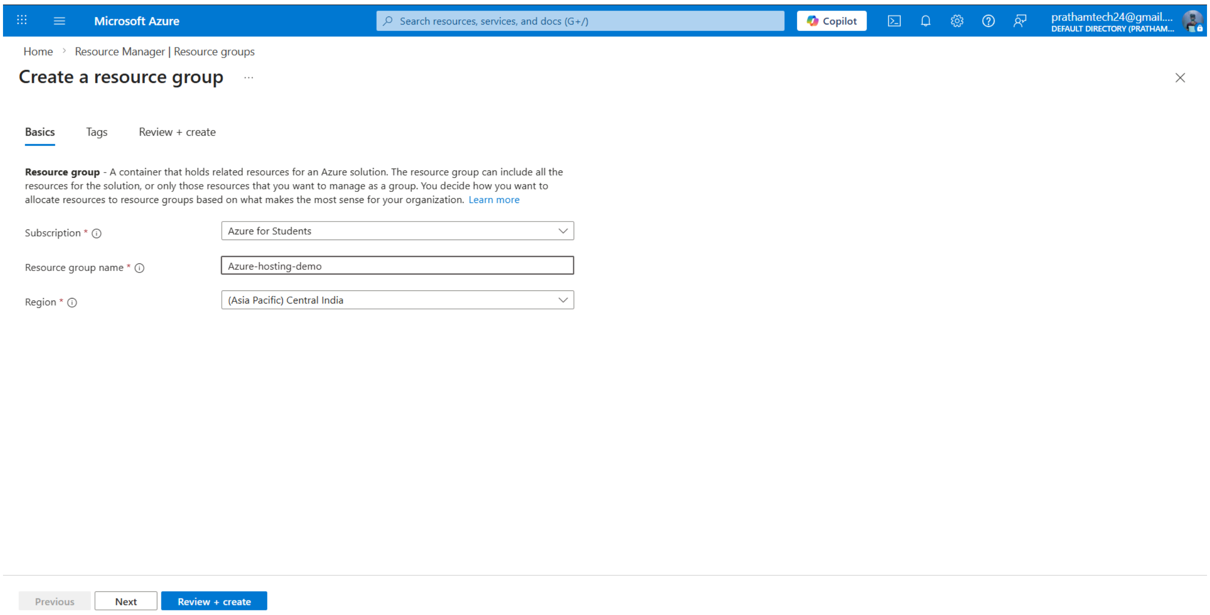
2. **Step:** Created an **Azure for Students** account.



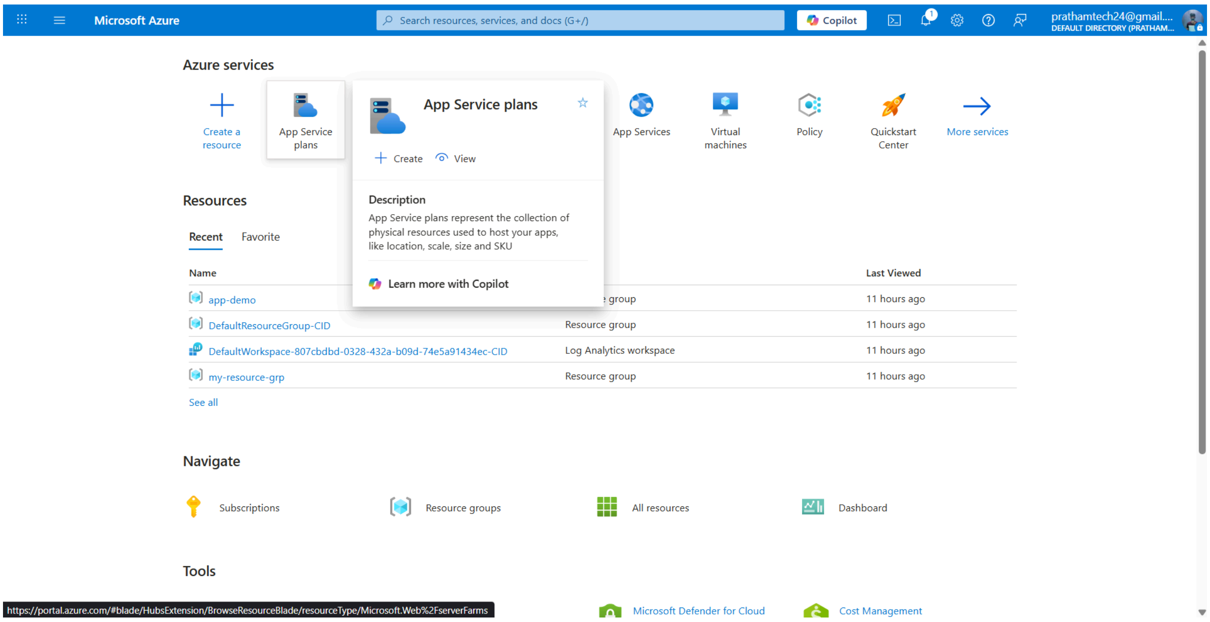


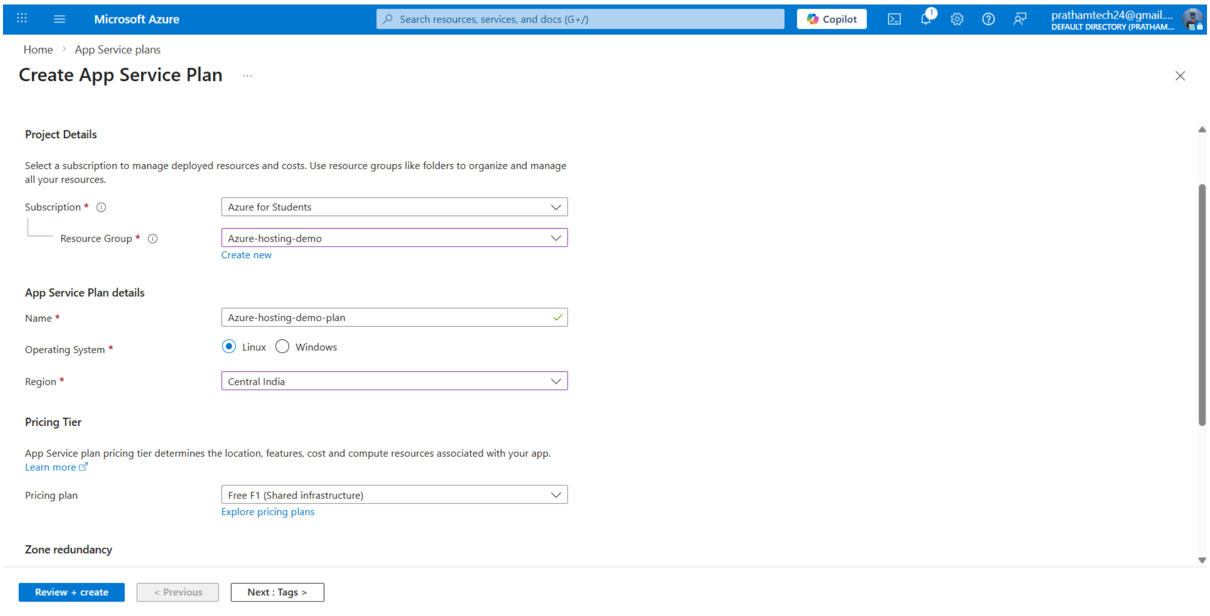
3. Resource Group (RG) Creation:



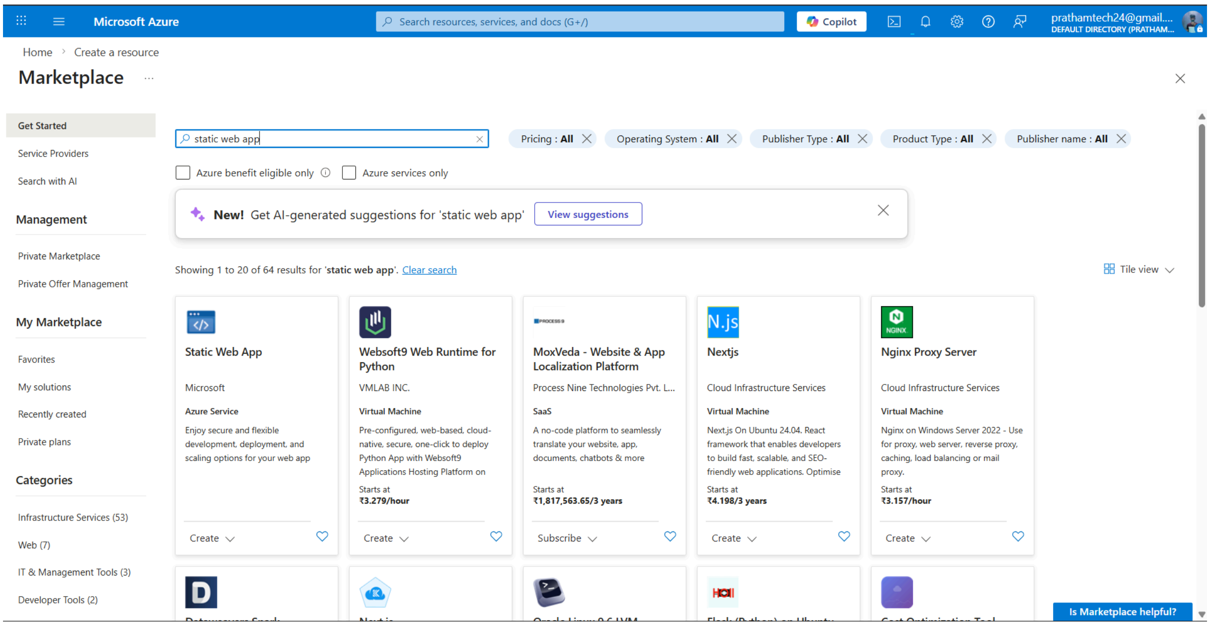


App Service Plan: Search **App Service Plans Create.**

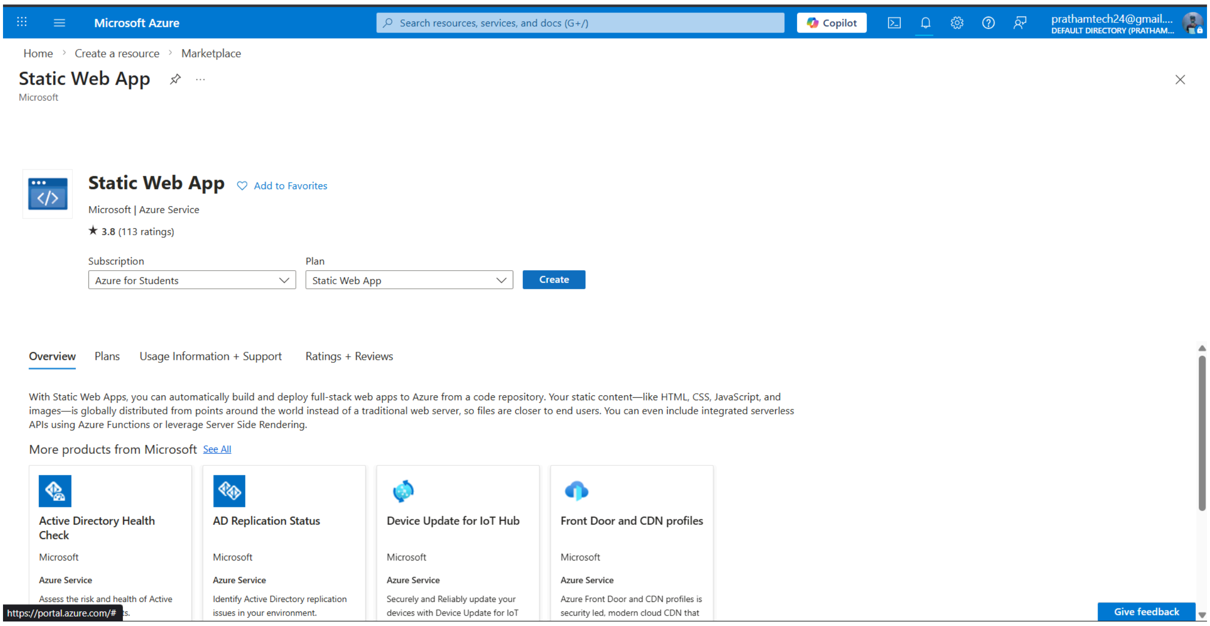




Search **Static Web Apps**

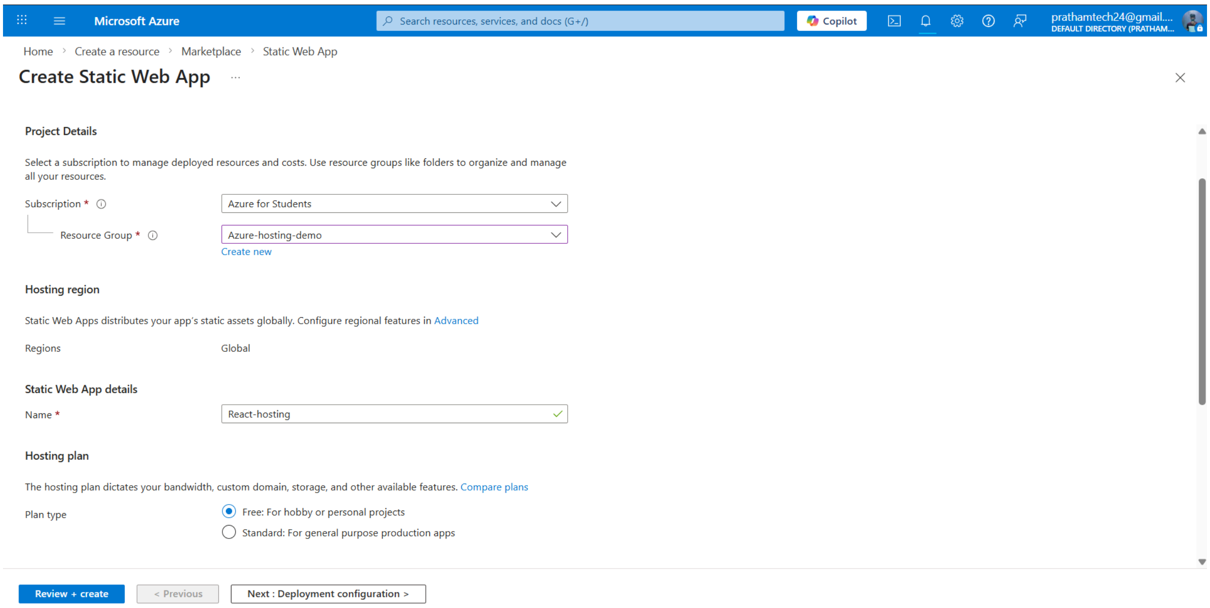


Create.



Basics Tab:

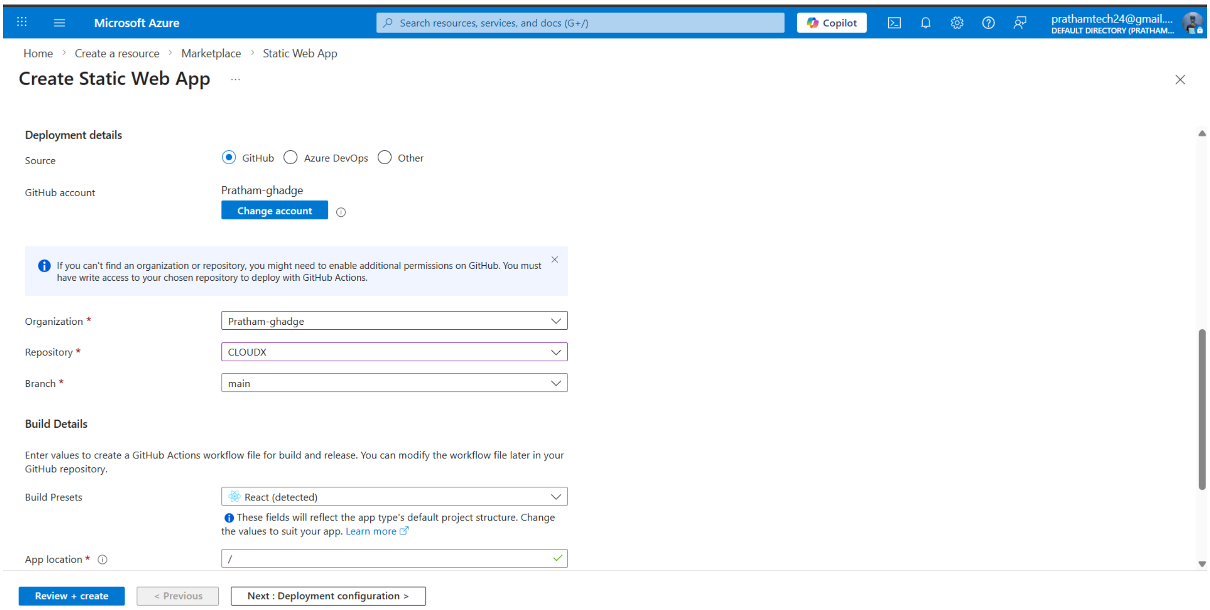
- **Subscription:** Azure for Students.
- **Resource Group:** Select Azure-hosting-demo.
- **Name:** React-hosting (or your project name).
- **Plan Type: Free** (For hobby/student projects).

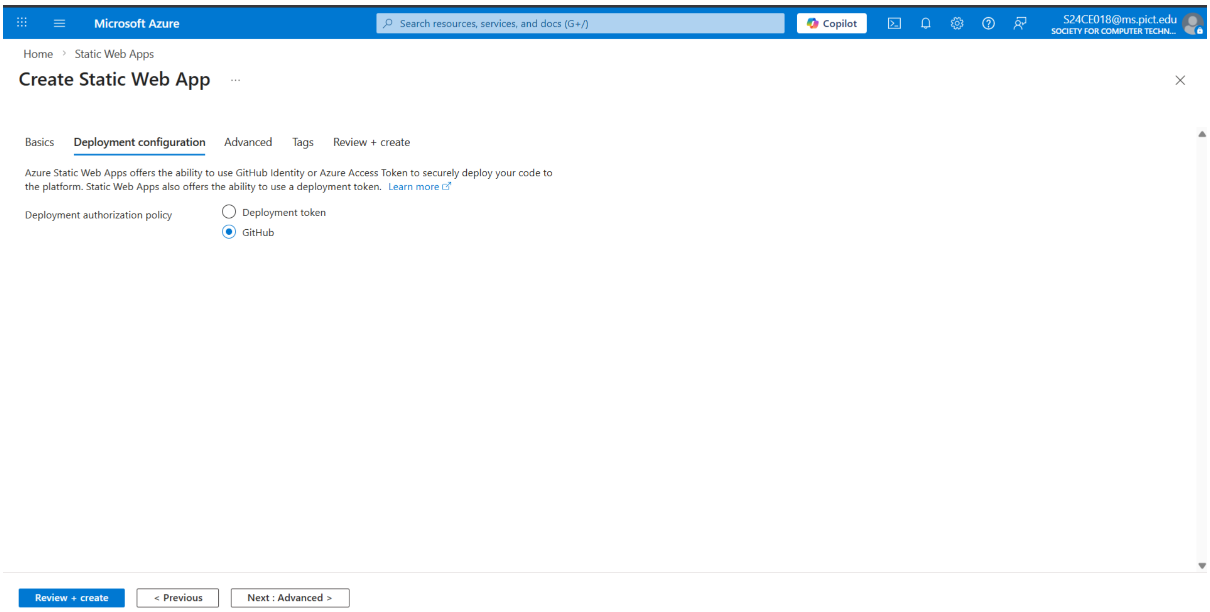


Select **GitHub**. **Sign in with GitHub**.

Authorize Azure to access your repositories.

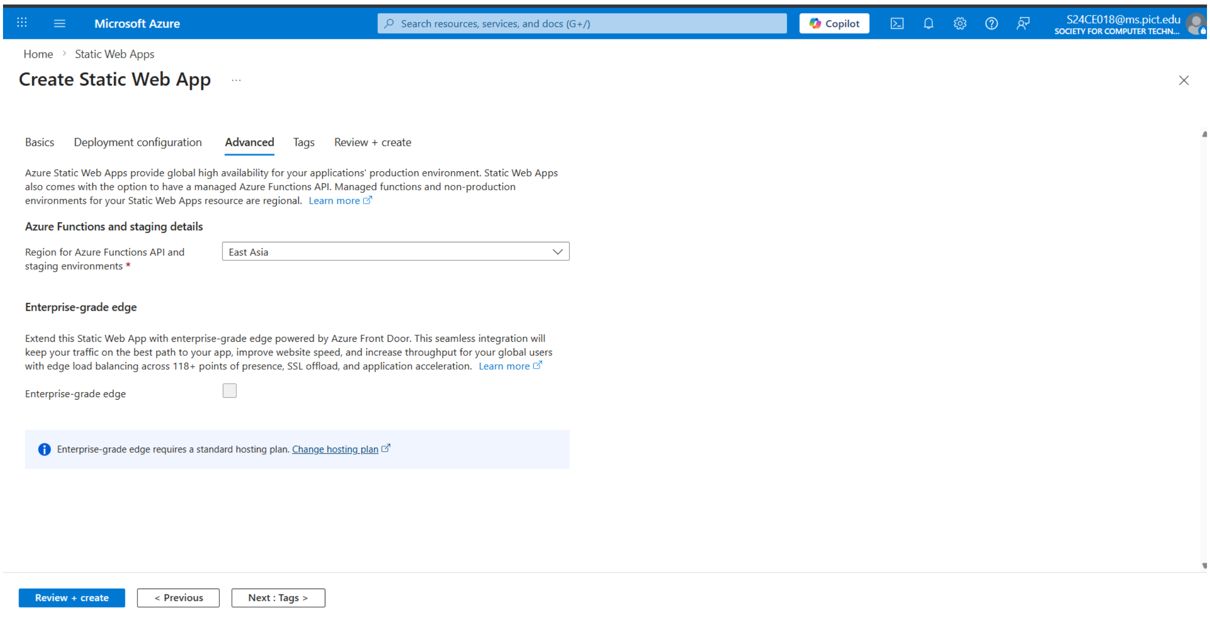
Select your **Organization**, **Repository**, and **Branch** (usually main).



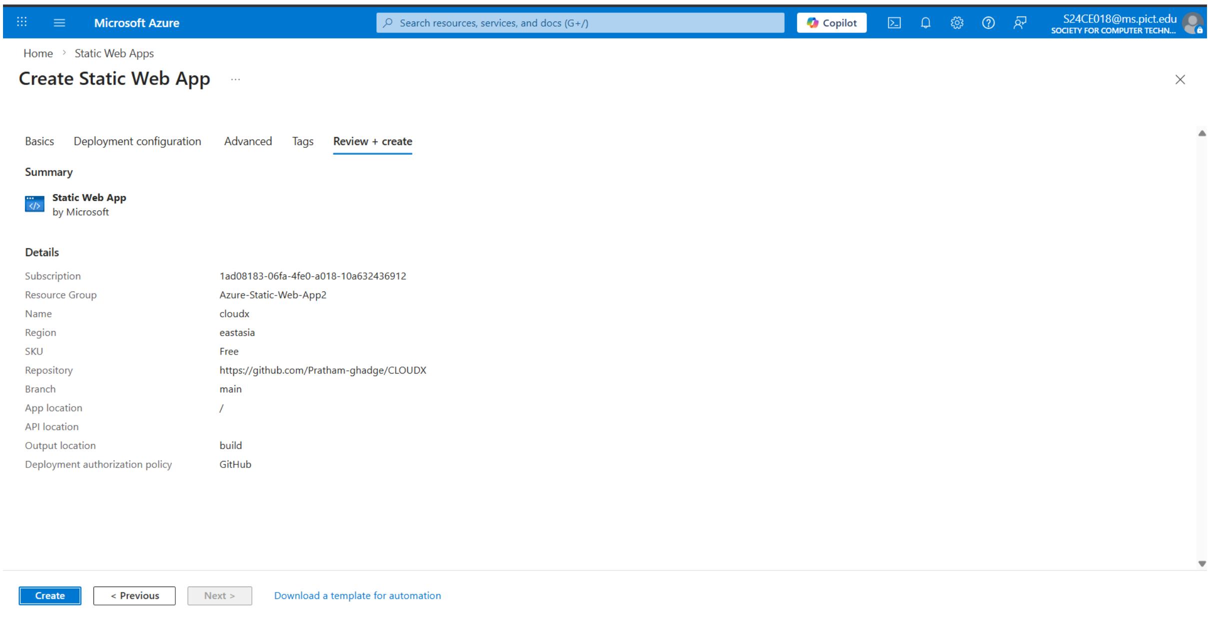


Advanced Networking:

- Check **Allowed Region Policy** to ensure compliance with Azure's student sandbox restrictions.



Finalize: Click **Review + Create**.



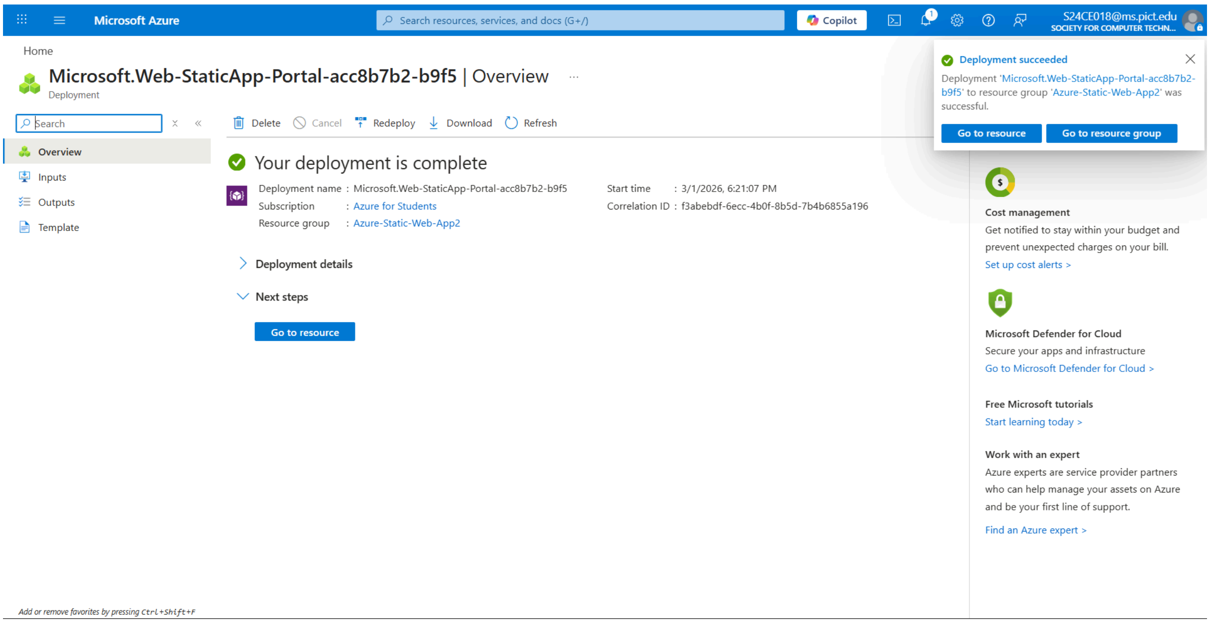
Phase 4: Automation & Deployment (The "Trigger")

Once created, Azure automatically injects a workflow file into your GitHub repository.

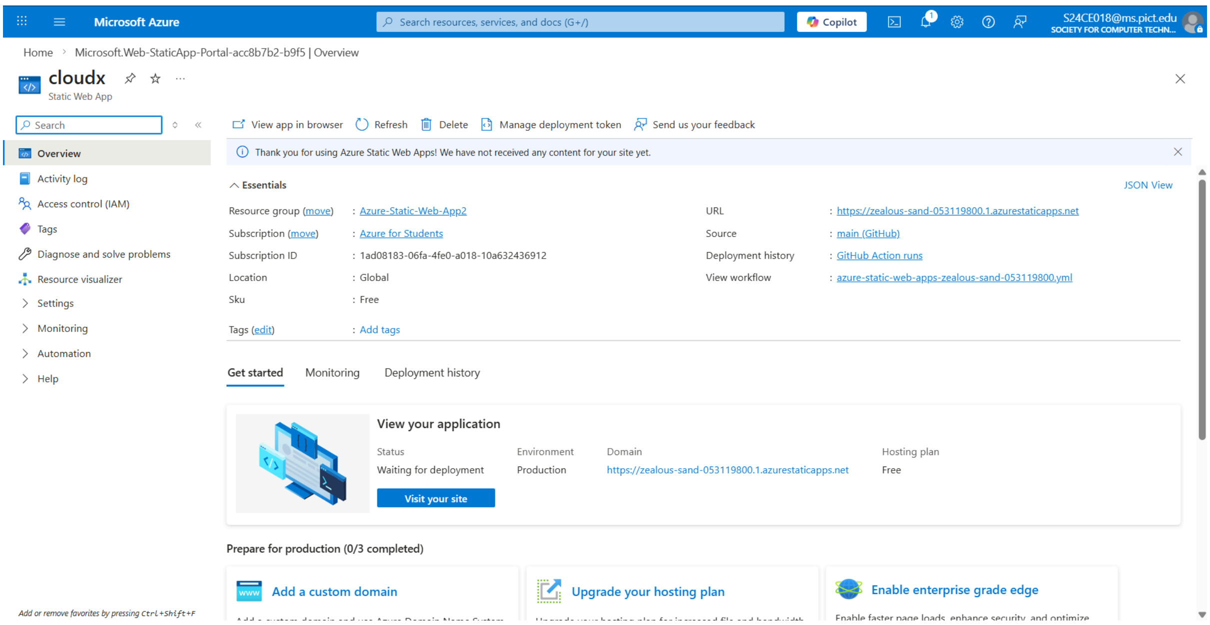
1. **GitHub Actions:** Go to your GitHub Repository **Actions** tab.
2. **Workflow Execution:** You will see a live "Job" running. This is the **CI/CD pipeline** compiling your code and pushing it to Azure servers.
3. **Deployment Success:** Once the green checkmark appears in GitHub, return to the Azure Portal.

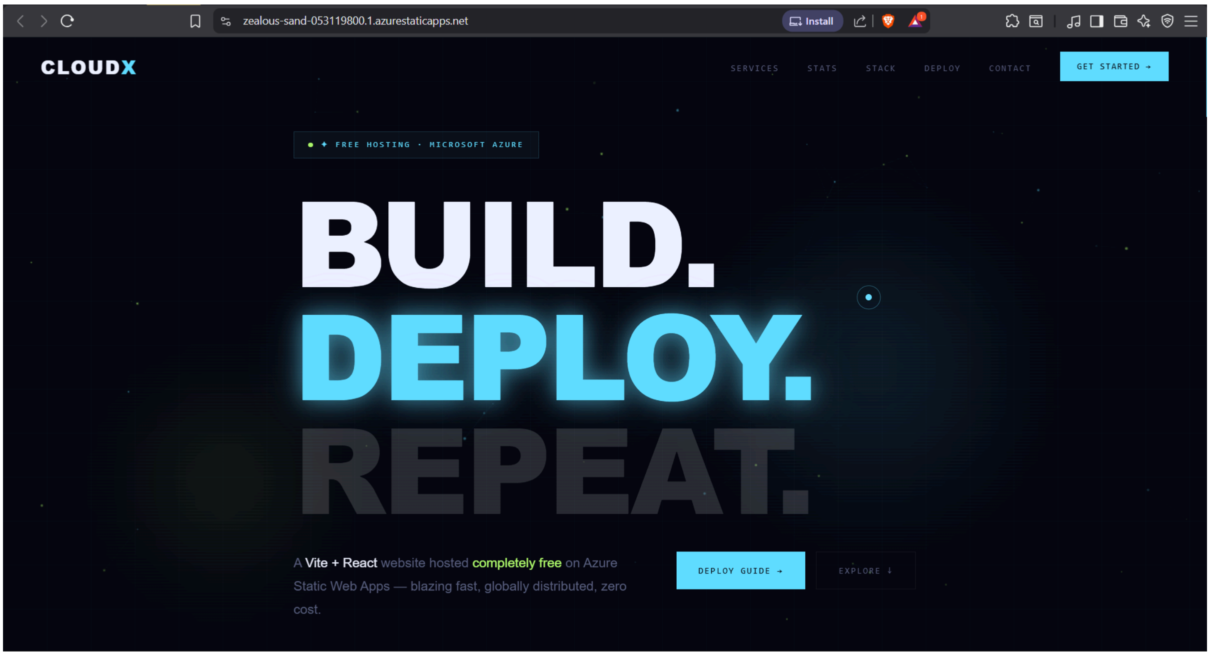
Once Azure displays the "Deployment Succeeded" message, follow these steps to verify the cloud environment.

1. **Resource Navigation:** Click the **Go to resource** button to open the Static Web App Management dashboard.



Live Access: Click the URL link. This will open your website in a new browser tab.





Conclusion :

By utilizing the **Azure Student Developer Pack**, this project successfully demonstrates a modern **DevOps workflow**. By decoupling the frontend hosting from a traditional server and using a **CI/CD pipeline**, we ensure that the deployment process is bug-free, automated, and highly scalable for future enhancements.